

System Architecture



Frontend

Component	Technology
Language Stack	HTML5 + CSS3 + JavaScript ES5/ES6 (Vanilla — no framework, no bundler)
Build Tool	None — static HTML files served directly by Express.js
Styling	css/style.css global sheet + css/animations.css + per-portal inline <style>
Routing	Browser navigation + JavaScript DOM show/hide panels
State Management	localStorage (shared across tabs) + sessionStorage (per-tab temp data)
3D Graphics	Three.js r128 — WebGL globe on portals & landing page (CDN)
Charts	Chart.js 4 — bar, doughnut, line charts in dashboards (CDN)
Icons	Font Awesome 6 (CDN)
Typography	Google Fonts: Poppins, Inter, Plus Jakarta Sans
Real-time Sync	window.storage event — cross-tab portal sync without WebSocket polling
Auth Persistence	JWT in localStorage; session auto-restored on every page load
AI Chat Widget	js/ecobot-widget.js — floating EcoBot panel, calls OpenAI via backend

Backend

Component	Package / Version
Runtime	Node.js >= 18.0.0
Framework	express ^4.18.2
ORM	prisma ^5.7.0 + @prisma/client ^5.7.0
Database	PostgreSQL (any modern version)
Real-time	socket.io ^4.6.1
Auth — tokens	jsonwebtoken ^9.0.2
Auth — passwords	bcryptjs ^2.4.3
File uploads	multer ^1.4.5 / multer-s3 ^3.0.1 (AWS S3)
Cloud storage	aws-sdk ^2.1490.0
AI integration	openai ^4.20.1
PDF generation	pdfkit ^0.14.0 + pdfmake ^0.2.9
QR codes	qrcode ^1.5.3
Payments	razorpay ^2.9.2
Email	nodemailer ^6.9.7
SMS	twilio ^4.19.0
Job queues	bull ^4.11.5 + ioredis ^5.3.2
HTTP security	helmet ^7.1.0 + cors ^2.8.5 + express-rate-limit ^7.1.5
Validation	joi ^17.11.0 + zod ^3.22.4 + express-validator ^7.0.1
Logging	winston ^3.11.0 + pino ^8.16.2 + morgan ^1.10.0

Component	Package / Version
Export	exceljs ^4.3.0 + xlsx ^0.18.5 + csv-parser ^3.0.0
API docs	swagger-jsdoc ^6.2.8 + swagger-ui-express ^5.0.0
Utilities	uuid ^9.0.0, compression ^1.7.4, dotenv ^16.3.1, chart.js ^4.4.0

Portal Architecture

Portal	HTML File	JS Connect File	Responsibilities
Organization Portal	org-portal.html	js/org-connect.js	ESG tracking, report gen, lab submission, monitoring
Laboratory Portal	lab-portal.html	js/lab-connect.js	Review submissions, approve/reject, NABL certs
Regulatory Portal	reg-portal.html	js/reg-connect.js	Gov review, certificate issuance, compliance notices
Landing Page	index.html	inline JS	Feature showcase, contact form, portal links
EcoBot Workspace	eco-bot-workspace.html	js/ecobot-widget.js	AI assistant (auth-gated, no direct links from portals)
Portal Select	portal-select.html	inline JS	Role-based router to correct portal

Cross-Portal Real-time Data Flow

Step	Actor	Action	localStorage Key / Event
1	Organization	Submit environmental report	eco_pending_lab_reports written
2	Lab Portal	Detect new submission	window.storage event fires
3	Lab Admin	Approve or Reject report	eco_lab_feedback + eco_reg_submissions
4	Reg Portal	Detect lab-approved report	window.storage on eco_reg_submissions
5	Reg Officer	Final review — approve/reject	eco_reg_submissions status updated
6	System	Issue digital certificate	eco_issued_certs + eco_reg_certs
7	Organization	Receive outcome notification	eco_notifications updated

Database Architecture

Component	Detail
Primary DB	PostgreSQL
ORM	Prisma v5 — schema-first, type-safe query builder
Schema	backend/prisma/schema.prisma
Migrations	backend/prisma/migrations/
Seed Script	backend/prisma/seed.js
Dev Database	embedded-postgres (npm dev dep) — in-process Postgres, no install needed
Cross-portal state	localStorage: eco_all_orgs, eco_pending_lab_reports, eco_lab_feedback, eco_reg_submissions, eco_issued_certs, eco_notifications, eco_contact_queries
Real-time sync	window.storage event fires in other browser tabs on same origin

Authentication & Security

Component	Detail
Token strategy	JWT — short-lived access token + long-lived refresh token in localStorage
Password storage	bcryptjs (bcrypt algorithm)
Roles	ENV_OFFICER, ADMIN, ENV_ENGINEER, PRODUCTION_HEAD, QUALITY_HEAD, HR_HEAD, LAB_ADMIN, REG_OFFICER
Rate limiting	express-rate-limit v7 on auth and API endpoints
HTTP security	helmet v7 — X-Frame-Options, CSP, HSTS, X-Content-Type-Options
CORS	cors package — origin whitelist configured per deployment
Input validation	Three layers: express-validator + joi + zod
Portal auth gate	Each portal checks token on load; browser back button auto-restores session
Security note	EcoBot SVG badges in portals must be non-clickable <g> groups, NOT <a href> (prevents auth bypass)

Compliance Standards Supported

Standard	Relevance
ISO 14001	Environmental Management Systems — core env compliance reporting standard
ISO 17025	Laboratory competence — basis for NABL lab portal requirements
GRI	Global Reporting Initiative — ESG scoring & sustainability reports
NABL	National Accreditation Board for Testing & Calibration Laboratories
CPCB	Central Pollution Control Board guidelines for industrial reporting
SEBI BRSR	Business Responsibility & Sustainability Report — India listed company ESG
Net Zero	Scope 1/2/3 emission accounting, Net Zero pathway, carbon credit management

Developer Tools & Scripts

Script / Tool	Description
npm run dev	nodemon src/server.js — auto-restart on file save
npm start	node src/server.js — production start
npm run prisma:migrate	Deploy pending database migrations
npm run prisma:seed	Populate database with initial/demo data
npm run prisma:studio	Open Prisma Studio — visual database browser
npm run prisma:gen	Regenerate Prisma Client after schema changes
npm test	Jest test suite (--runInBand --forceExit)
npm run lint	ESLint check on src/ directory
npm run swagger	Generate Swagger/OpenAPI documentation
nodemon ^3.0.2	Dev dependency — auto-restart
jest ^29.7.0	Dev dependency — test runner
supertest ^6.3.3	Dev dependency — HTTP API testing
eslint ^8.55.0	Dev dependency — linter
prisma ^5.7.0	Dev dependency — CLI for migrations & studio
embedded-postgres ^18.4.0-beta.17	Dev dependency — in-process PostgreSQL

Environment Variables

Variable	Required	Purpose
DATABASE_URL	Required	PostgreSQL connection string
JWT_SECRET	Required	Secret key for JWT access tokens
JWT_REFRESH_SECRET	Required	Secret key for JWT refresh tokens
PORT	Optional	Server port (default 5000)
OPENAI_API_KEY	Optional	OpenAI API key for EcoBot AI
AWS_ACCESS_KEY_ID	Optional	AWS credentials for S3
AWS_SECRET_ACCESS_KEY	Optional	AWS credentials for S3
AWS_S3_BUCKET	Optional	S3 bucket name
AWS_REGION	Optional	AWS region (e.g. ap-south-1)
REDIS_URL	Optional	Redis URL for Bull job queues
RAZORPAY_KEY_ID	Optional	Razorpay payment gateway key
RAZORPAY_KEY_SECRET	Optional	Razorpay payment gateway secret
TWILIO_ACCOUNT_SID	Optional	Twilio account SID for SMS
TWILIO_AUTH_TOKEN	Optional	Twilio auth token
TWILIO_PHONE_NUMBER	Optional	Twilio sender number
SMTP_HOST	Optional	Email SMTP server
SMTP_USER	Optional	SMTP username
SMTP_PASS	Optional	SMTP password
SMTP_FROM	Optional	From address for emails